

Ex 1: $f(x) = \frac{2x}{\theta^2} 1_{[0, \theta]}(x)$

$$E(X) = \int_{\mathbb{R}} x f(x) dx$$

$$= \int_0^{\theta} x \frac{2x}{\theta^2} dx$$

$$= \frac{2}{\theta^2} \int_0^{\theta} x^2 dx$$

$$= \frac{2}{\theta^2} \left[\frac{1}{3} x^3 \right]_0^{\theta}$$

$$= \frac{2}{\theta^2} \cdot \frac{\theta^3}{3} = \frac{2}{3} \theta$$

Ex 2

$$1_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

$$E(X^2) = \int_{\mathbb{R}} x^2 f(x) dx$$

$$= \frac{2}{\theta^2} \int_0^{\theta} x^3 dx$$

$$= \frac{2}{\theta^2} \left[\frac{1}{4} x^4 \right]_0^{\theta}$$

$$= \frac{2}{\theta^2} \cdot \frac{1}{4} \theta^4 = \frac{1}{2} \theta^2$$

$$V(X) = E(X^2) - E^2(X)$$

$$= \frac{\theta^2}{2} - \left(\frac{2}{3} \theta \right)^2$$

$$= \frac{\theta^2}{2} - \frac{4}{9} \theta^2$$

$$= \frac{9\theta^2 - 8\theta^2}{18}$$

$$= \frac{\theta^2}{18}$$

$$P\left(X < \frac{\theta}{4}\right) = \int_{-\infty}^{\frac{\theta}{4}} f(x) dx$$

$$= \int_0^{\frac{\theta}{4}} \frac{2x}{\theta^2} 1_{[0, \theta]} dx$$

$$= \frac{2}{\theta^2} \int_0^{\frac{\theta}{4}} x dx = \frac{2}{\theta^2} \left[\frac{1}{2} x^2 \right]_0^{\frac{\theta}{4}}$$

$$= \frac{2}{\theta^2} \cdot \frac{1}{2} \cdot \frac{\theta^2}{16}$$

$$= \frac{1}{16}$$

Ex 2: $X \sim Y$ joint index

$$1) \int_{\mathbb{R}} f(t) dt = 1$$

$$\Leftrightarrow \int_{\mathbb{R}} t e^{-t} 1_{[0, +\infty[} dt = 1$$

$$\Leftrightarrow \int_0^{+\infty} t e^{-t} dt$$

compos: $U = t \rightarrow U' = 1$

$$V' = e^{-t} \rightarrow V = -e^{-t}$$

$$\Rightarrow \left[-t e^{-t} \right]_0^{+\infty} + \int_0^{+\infty} e^{-t} dt \rightarrow$$

$$\Leftrightarrow C \int_0^{+\infty} e^{-t} dt = 1$$

$$\Leftrightarrow C = 1$$

$$2) E(Y) = \int_{\mathbb{R}} y f(y) dy$$

$$= \int_0^{+\infty} y^2 e^{-y} dy$$

$$E(y^2) = \underbrace{\int_0^{+\infty} y^3 e^{-y} dy}_0 + \underbrace{3 \int_0^{+\infty} y^2 e^{-y} dy}_2$$

$$= 3 \times 2$$

$$= 6$$

$$V(y) = E(y^2) - E^2(y)$$

$$= 6 - (2^2) = 2$$

$$3) P(y < 4 / y > 1)$$

$$= P(y < 4, y > 1)$$

$$P(y > 1)$$

$$= \frac{P(1 < y < 4)}{P(y > 1)}$$

$$P(1 < y < 4) = \int_1^4 y e^{-y} dy$$

$$= \int_1^4 y e^{-y}$$

$$= [-y e^{-y}]_1^4 + [-e^{-y}]_1^4$$

$$= -4e^{-4} + e^{-1} - e^{-4} + e^{-1}$$

$$= -5e^{-4} + 2e^{-1}$$

$$\begin{aligned}
 &= \begin{cases} 0 & \text{if } y \leq 0 \\ \int_0^y t e^{-t} dt & \text{if } y > 0 \end{cases} \\
 &= \begin{cases} 0 & \text{if } y \leq 0 \\ \left[-te^{-t} \right]_0^y + \left[-e^{-t} \right]_0^y & \text{if } y > 0 \end{cases} \\
 &= \begin{cases} 0 & \text{if } y \leq 0 \\ -ye^{-y} + 0e^{-0} & \text{if } y > 0 \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 &= \begin{cases} 0 & \text{if } y \leq 0 \\ (-ye^{-y} - e^{-y} + 1 \sin y) & \text{if } y > 0 \end{cases} \\
 &= \begin{cases} 0 & \text{if } y \leq 0 \\ 1 - e^{-y}(y+1) & \text{if } y > 0 \end{cases} \\
 &= F_Y(y)
 \end{aligned}$$

$$\begin{aligned}
 F_Z(z) &= F_Y(z^2) \\
 &= \begin{cases} 1 - e^{-z^2}(z^2+1) & \text{if } z^2 > 0 \\ 0 & \text{if } z^2 \leq 0 \end{cases}
 \end{aligned}$$

$$\Rightarrow f_Z(z) = \left(-2ze^{-z^2}(z^2+1) + 2ze^{-z^2} \right)$$

$$\begin{aligned}
 &= \left[-2ze^{-z^2} \right] \\
 &= 2z^3 e^{-z^2} \quad (11_R) \\
 f_Z(z) &= 2z^3 e^{-z^2}
 \end{aligned}$$

exercice 3

$$\text{Soit } f_X(x) = \frac{1}{2} [1 - x]^{-1} 1_{(0,1)}(x)$$

$$\text{Soit } Y = e^X$$

$$\begin{aligned} F_Y(y) &= P\left(Y \leq \frac{y}{2}\right) = P(e^X \leq y) \\ &= P(X \leq \ln(y)) \end{aligned}$$

$$= F_X(\ln(y))$$

$$f_Y\left(\frac{y}{2}\right) = \left(f_X(\ln(y))\right)'$$

$$= \ln\left(\frac{y}{2}\right) \times f_X(\ln(y))$$

$$= \frac{1}{2} \times \frac{1}{2} [1 - \ln(y)]^{-1} 1_{(0,1)}(\ln(y))$$

$$= \frac{1}{2y} \times [e^{-1}, e^1](y)$$

$$\begin{aligned}
 E(y) &= \int_{\mathbb{R}} y f(y) \\
 &= \int_{e^{-1}}^{e^1} \frac{1}{2y} y dy \\
 &= \int_{e^{-1}}^{e^1} \frac{1}{2} dy \\
 &= \frac{1}{2} (e^1 - e^{-1})
 \end{aligned}$$

$$\begin{aligned}
 E(y^2) &= \int_{e^{-1}}^{e^1} \frac{y}{2} dy \\
 &= \frac{1}{4} [y^2]_{e^{-1}}^{e^1} \\
 &= \frac{1}{4} (e^2 - e^{-2})
 \end{aligned}$$

$$\begin{aligned}
 V(y) &= \frac{1}{4} (e^2 - e^{-2}) - \frac{1}{4} (e^1 - e^{-1})^2 \\
 &= \frac{1}{4} (e^2 - e^{-2} - e^2 - e^{-2} + 2e) \\
 &= \frac{1}{4} (-2e^{-2} + 2) \\
 &= \frac{1}{2} - \frac{e^{-2}}{2}
 \end{aligned}$$

exercice 4:

$$y = x$$

$$1) x \sim \mathcal{U}([-1, 1])$$

$$y = g(x)$$

$$\Leftrightarrow x = g^{-1}(y)$$

$$f_Y(y) = f_X(g^{-1}(y)) \times |(g^{-1}(y))'|$$

$$y = x^2 \Rightarrow g(x) = x^2$$

$$g(x) = y$$

$$\Leftrightarrow x^2 = y$$
$$\Leftrightarrow x = \begin{cases} -\sqrt{y} \\ \sqrt{y} \end{cases}$$

$$f_Y(y) = f_X(-\sqrt{y}) \times |(-\sqrt{y})'| + f_X(\sqrt{y}) \times |(\sqrt{y})'|$$

$$= \frac{1}{2} \mathbb{1}_{\mathbb{L}_{[-1, 1]}}(-\sqrt{y}) \times \left| -\frac{1}{2\sqrt{y}} \right| + \frac{1}{2} \mathbb{1}_{\mathbb{L}_{[-1, 1]}}(\sqrt{y}) \times \left| \frac{1}{2\sqrt{y}} \right|$$

$$= \frac{1}{4\sqrt{y}} \left(\mathbb{1}_{\mathbb{L}_{[-1, 1]}}(-\sqrt{y}) + \mathbb{1}_{\mathbb{L}_{[-1, 1]}}(\sqrt{y}) \right)$$

$$-1 \leq \sqrt{y} \leq 1$$

$$\Rightarrow 0 \leq y \leq 1$$

$$f_Y(y) = \frac{1}{2\sqrt{y}} \left[\frac{1}{\sqrt{2\pi}} e^{-\frac{(-\sqrt{y})^2}{2\sigma^2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{(\sqrt{y})^2}{2\sigma^2}} \right]$$

$$= \frac{1}{2\sqrt{y}} \frac{1}{\sqrt{2\pi}} \left(e^{-\frac{y}{2\sigma^2}} + e^{-\frac{y}{2\sigma^2}} \right)$$

$$2. \quad X \sim N(m, \sigma^2)$$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

$$f_Y(y) = \frac{1}{2\sqrt{y}} \left(f_X(-\sqrt{y}) + f_X(\sqrt{y}) \right)$$

$$= \frac{1}{2\sqrt{y}} \times \frac{1}{\sigma\sqrt{2\pi}} \left[e^{-\frac{(-\sqrt{y}-m)^2}{2\sigma^2}} + e^{-\frac{(\sqrt{y}-m)^2}{2\sigma^2}} \right]$$

$$= \frac{1}{2\sigma\sqrt{2\pi y}} \left(e^{-\frac{(\sqrt{y}+m)^2}{2\sigma^2}} + e^{-\frac{(\sqrt{y}-m)^2}{2\sigma^2}} \right)$$

exercice 6:

X et Y sont indep

$$X \sim \mathcal{E}(\lambda)$$

$$Y \sim \mathcal{E}(\mu)$$

$$f_X(x) = \lambda e^{-\lambda x} \quad \lambda > 0, x \geq 0$$

$$f_Y(y) = \mu e^{-\mu y} \quad \mu > 0, y \geq 0$$

$$U = \min(X, Y)$$

$$F_U(u) = \mathbb{P}(U < u) = 1 - \mathbb{P}(U \geq u)$$

$$= 1 - \mathbb{P}(\min(X, Y) \geq u)$$

$$= 1 - \mathbb{P}(X \geq u, Y \geq u)$$

$$= 1 - \mathbb{P}(X \geq u) \times \mathbb{P}(Y \geq u)$$

$$= 1 - (1 - F_X(u))(1 - F_Y(u))$$

$$F_X(x) = \mathbb{P}(X \leq x)$$

$$= \int_{-\infty}^x f_X(t) dt$$

$$= \int_{-\infty}^x 1 \cdot e^{-1 \cdot t} \mathbb{1}_{[0, +\infty)}(t) dt$$

$$= \begin{cases} 0 & \text{if } x \leq 0 \\ \int_0^x 1 \cdot e^{-1 \cdot t} dt & \text{if } x > 0 \end{cases}$$

$$= \begin{cases} 0 & \text{if } x \leq 0 \\ [-e^{-1 \cdot t}]_0^x & \text{if } x > 0 \end{cases}$$

$$= \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-1 \cdot x} & \text{if } x > 0 \end{cases}$$

$$F_Y(y) = 1 - (1 - F_X(y))(1 - F_Y(y))$$

$$= 1 - e^{-1 \cdot y} \times e^{-\mu y} = 1 - e^{-y(1+\mu)}$$

$$u \sim \mathcal{E}(1+\mu)$$

$\Rightarrow$